

Volume II · Chapter 6 — Transformer Architecture & Self-Attention · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-6-context.md for the AI interaction log.

Prerequisites: Ch 4 (ML), Ch 5 (NLP basics).

Learning outcomes — the student can: explain self-attention and the transformer block; describe tokenization, embeddings and positional encoding; distinguish encoder/decoder/encoder-decoder families; read a model card and reason about parameters/context length.

Topics (theory) — 12 h:

#	Topic	h
6.1	From embeddings to context; limits of classical NLP	1
6.2	Tokenization & subwords (BPE, WordPiece)	2
6.3	Self-attention & multi-head attention	3
6.4	The transformer block: FFN, residuals, layer norm, positional encoding	2
6.5	Architectures: encoder (BERT), decoder	2

#	Topic	h
	(GPT), encoder-decoder (T5)	
6.6	Pretraining objectives (MLM, causal LM) & scaling intuition	1
6.7	Reading model cards: parameters, context length, tokens	1

Datasets/tools: Hugging Face transformers; a small BERT/GPT; clinical text samples. **Assessment:** architecture quiz (**50%**); attention-visualization writeup (**30%**); tokenization exercise (**20%**). **Key decisions:** encoder vs. decoder for a task; context length vs. cost; tokenizer choice. **References:** plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory **12** + Lab **6** = **18**.